

Adversarial Perturbations in Convolutional Neural Networks: A Geometric Analysis

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September 4, 2026

Abstract

This report investigates the geometric fragility of Convolutional Neural Networks (CNNs). I trained a VGG-style CNN (Simonyan & Zisserman, 2014) on the CIFAR-10 dataset (Krizhevsky, 2009) to establish a baseline decision boundary, which I then attacked using the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD). The results show that adversarial vulnerability follows directly from the piecewise-linear structure of ReLU networks, rather than from software bugs (Goodfellow et al., 2015).

1 Introduction & Task Formulation

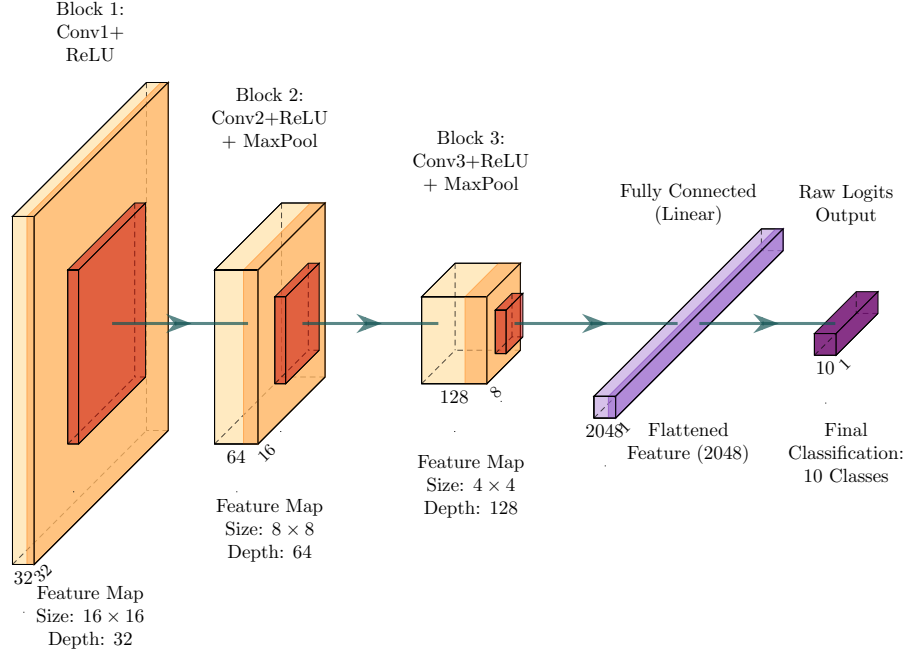
This project builds on the CIFAR-10 classification lab from Week 10. CNNs trained to high accuracy on clean images can be misclassified by small, targeted pixel perturbations; the question is whether this follows from a structural property of the model. To test if this vulnerability stems from the linearity of ReLU boundaries, I designed a CNN, applied FGSM and PGD attacks, and extracted the geometric alignment of the gradients during the attack loop.

2 CNN Framework & Architecture

The architecture was constrained to train within the memory and time budget of a Colab T4 GPU. LeNet-5 was initially considered, however it lacks Batch Normalisation, making it poorly suited to PGD attacks. Conversely, models like ResNet-50 were too computationally expensive for the Colab T4 environment. I designed a VGG-style architecture (Simonyan & Zisserman, 2014) using uniform 3×3 filters, which ensures spatially uniform receptive fields and makes the gradient flow through the network geometrically predictable. The resulting dimension mapping is $\mathbb{R}^{32 \times 32 \times 3} \rightarrow \mathbb{R}^{16 \times 16 \times 32} \rightarrow \mathbb{R}^{8 \times 8 \times 64} \rightarrow \mathbb{R}^{4 \times 4 \times 128}$.

For activation, I used Rectified Linear Units (ReLU). ReLU's linear regions allow the gradient to point in a consistent direction (Montufar et al., 2014), enabling sign-based attacks.

I applied Batch Normalisation after each convolution. Santurkar et al. (2018) showed that Batch Normalisation restricts the Lipschitz constant of the gradient; without it during testing, the loss oscillated heavily and PGD occasionally diverged. The network outputs raw logits $\mathbf{z} \in \mathbb{R}^{10}$; passing them through Softmax would suppress large-logit gradients (gradient masking), corrupting the attack signal.



Dimension Mapping: $R^{32 \times 32 \times 3} \rightarrow R^{16 \times 16 \times 32} \rightarrow R^{8 \times 8 \times 64} \rightarrow R^{4 \times 4 \times 128} \rightarrow 2048 \rightarrow 10$

Figure 1: VGG style CNN Architecture for CIFAR-10 classification.

3 Optimisation Dynamics & Baseline Training

I minimised the Cross Entropy Loss and added L_2 weight decay to prevent overfitting. The loss function is:

$$L(\theta) = -\frac{1}{N} \sum_{i=1}^N \log P(y_i | \mathbf{x}_i; \theta) + \lambda \|\theta\|_2^2 \quad (1)$$

Optimiser Selection. SGD is prone to oscillation in the non-convex Cross-Entropy loss surface; Adam (Kingma & Ba, 2014) applies per-parameter learning rates based on a running average of past gradients, which stabilises convergence.

I added a StepLR scheduler to reduce the learning rate by a factor of ten at epochs 10 and 20 (visible in Figure 5). Training converged to a 78.5% clean test accuracy.

4 Adversarial Attack Implementation

After training, I froze the model parameters θ and attacked the input pixels $\mathbf{x}$. Both attacks perturb $\mathbf{x}$ along $\nabla_{\mathbf{x}} J(\theta, \mathbf{x}, y)$ to maximise the loss.

Fast Gradient Sign Method (FGSM). FGSM (Goodfellow et al., 2015) assumes J is locally linear and takes a single step of size ϵ in the sign-gradient direction:

$$\mathbf{x}_{adv} = \mathbf{x} + \epsilon \cdot \text{sign}(\nabla_{\mathbf{x}} J(\theta, \mathbf{x}, y)) \quad (2)$$

Geometric Limitation: FGSM assumes J is locally linear. When J is non-linear within the ϵ -ball, the single-step update undershoots the loss maximum, as the first-order approximation becomes inaccurate.

Projected Gradient Descent (PGD). Madry et al. (2018) formulated PGD to take multiple smaller steps (α). The attack is initialised with uniform noise to avoid converging to the same local loss maximum on every run.

$$\mathbf{x}^{(0)} = \mathbf{x} + \mathcal{U}(-\epsilon, \epsilon) \quad (3)$$

The iterative step is:

$$\mathbf{x}^{(t+1)} = \Pi_{\mathcal{B}_\epsilon(\mathbf{x})} \left(\mathbf{x}^{(t)} + \alpha \cdot \text{sign}(\nabla_{\mathbf{x}} J(\theta, \mathbf{x}^{(t)}, y)) \right) \quad (4)$$

Geometric Constraint: The projection operator Π clamps the values, keeping $\|\delta\|_\infty \leq \epsilon$ and ensures the pixels remain in the valid $[0, 1]$ range.

5 Analysis of Results

To test whether Goodfellow’s linear hypothesis explains the observed vulnerability, I plotted the loss profile along the adversarial direction, the gradient norm trajectory over PGD iterations, and the perturbations directly.

5.1 Global Vulnerability: The Epsilon Sweep

Figure 2 shows accuracy versus ϵ . At $\epsilon = 0.03$, FGSM accuracy falls to 0.6%, while PGD falls to 0.0%. Madry et al. (2018) predicted this result: PGD finds stronger attacks than FGSM because it tracks the loss surface across multiple steps rather than relying on a single linear approximation.

5.2 The Geometric Paradox: Visual Triplets

Figure 6 shows the original, perturbation, and adversarial images. The min-max scaled perturbation reveals spatially non-uniform, gradient-aligned structure, not uniform noise. Contrarily, the adversarial image remains perceptually indistinguishable because $\|\delta\|_\infty$ is bounded to 0.05 (Szegedy et al., 2013). The network changes class when δ points along ∇J , because piecewise-linear ReLU boundaries require only directional alignment, not large perturbation magnitude, to cross a decision boundary.

5.3 High-Dimensional Anisotropy: 1D Loss Slice

Figure 4 compares loss along adversarial and random directions. In $\mathbb{R}^{3072}$, random vectors are nearly orthogonal to ∇J (Goodfellow et al., 2015). A random direction $\mathbf{v}$ with $\|\mathbf{v}\|_2 = \|\mathbf{g}\|_2$ produces $J(\mathbf{x} + \mathbf{v}) \approx J(\mathbf{x})$, consistent with the near-orthogonality argument. Conversely, stepping along the adversarial vector $\mathbf{g}$ pushes the Cross-Entropy loss to ~ 65 at $\alpha = 1$. Since $\|\mathbf{v}\|_2 = \|\mathbf{g}\|_2$, the asymmetry in loss ascent is attributable entirely to alignment with ∇J .

5.4 Local Optimisation Trajectory: Gradient Norms

Figure 3 tracks $\|\nabla_{\mathbf{x}} J\|_2$ over 40 PGD iterations. The norm jumps to ~ 42 by step 2. After step 3, the norm oscillates because the L_∞ clamping prevents further ascent in those dimensions.

5.5 Quantitative Verification of the Linear Hypothesis

I computed cosine similarity and Taylor error for one correctly-classified sample (Class 0, Airplane).

Cosine similarity $\cos(\boldsymbol{\delta}, \nabla J)$ measures how well the attack aligns with the gradient. If ∇J were Gaussian, $\text{sign}(\nabla J)$ would align with it at $\sqrt{2/\pi} \approx 0.797$ on average. FGSM yields a cosine similarity of 0.7451, consistent with the $\sqrt{2/\pi}$ expectation under i.i.d. Gaussian gradient assumptions. The gap comes from the gradients not being purely Gaussian and from L_∞ clamping. PGD’s final alignment was lower: 0.5737. Each PGD step evaluates ∇J at a different $\mathbf{x}$, so the final $\boldsymbol{\delta}$ mixes directions. This is expected, as PGD leaves the strict local linear region after the first step.

The Taylor error measures how far J departs from linearity:

$$\text{Error} = \frac{|J(\mathbf{x} + \boldsymbol{\delta}) - J(\mathbf{x}) - \boldsymbol{\delta}^T \nabla_{\mathbf{x}} J|}{|J(\mathbf{x} + \boldsymbol{\delta}) - J(\mathbf{x})|} \quad (5)$$

The linear hypothesis gets the initial direction right, but the Taylor error is 0.9597 for FGSM and 0.9956 for PGD at $\epsilon = 0.05$. At small ϵ , $J \approx J(\mathbf{x}) + \boldsymbol{\delta}^T \nabla J$, but at $\epsilon = 0.05$, the higher-order terms dominate. An error of 0.9597 means FGSM’s linear prediction accounts for less than 5% of the true loss increase; iterative updates (PGD) are required because each step re-evaluates ∇J at the current $\mathbf{x}^{(t)}$ progressively tracking the non-linear loss surface.

6 Theoretical Defences & Conclusion

Several constraints limit how far these results generalise. First, the geometric statistics in Section 5.5 (cosine similarity, Taylor error) are computed on a single correctly-classified image; while the epsilon sweep in Section 5.1 uses the full 10,000-image test set, the fine-grained alignment analysis has not been averaged across classes or samples, so it should be read as an illustrative case study rather than a population-level estimate. Second, the architecture itself is intentionally small, three convolutional blocks trained to 78.5% clean accuracy chosen to fit within a Colab T4 budget rather than to maximise accuracy; results may not transfer directly to deeper or higher-capacity networks, where the loss landscape’s curvature could differ. Third, all attacks were evaluated under a white-box threat model with full gradient access; the geometric fragility observed here says nothing about robustness under black-box or transfer-attack settings, which rely on different (and often weaker) approximations of $\nabla_{\mathbf{x}} J$. Finally, PGD was run with a single random initialisation per image rather than multiple restarts, so the reported attack success rates are a lower bound on true worst-case vulnerability, and no adversarial training was actually implemented. Section 6’s min-max formulation is presented as a theoretical response to the geometric argument, not as an empirically tested defence. Standard training minimises loss on clean images, not adversarial ones. Madry et al. (2018) reformulated training as a min-max optimisation problem:

$$\min_{\theta} \mathbb{E}_{(\mathbf{x}, y)} \left[\max_{\boldsymbol{\delta} \in \mathcal{B}_\epsilon} J(\theta, \mathbf{x} + \boldsymbol{\delta}, y) \right] \quad (6)$$

Each update uses the PGD attack on the current batch. Carlini & Wagner (2017) demonstrated that purported defences are only meaningful when evaluated against the strongest adaptive adversary, not fixed attacks.

Until adversarial training becomes standard practice, gradient-based attacks will continue to bypass CNNs.

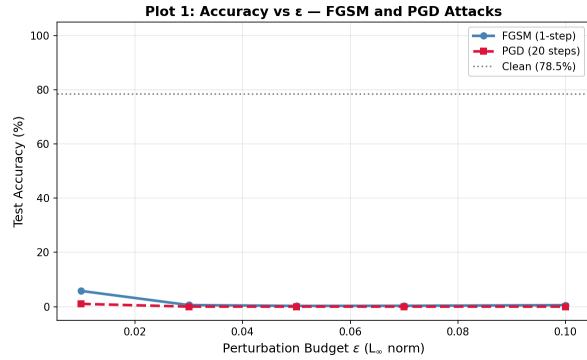


Figure 2: Accuracy decay curve evaluating $\epsilon \in [0.01, 0.1]$.

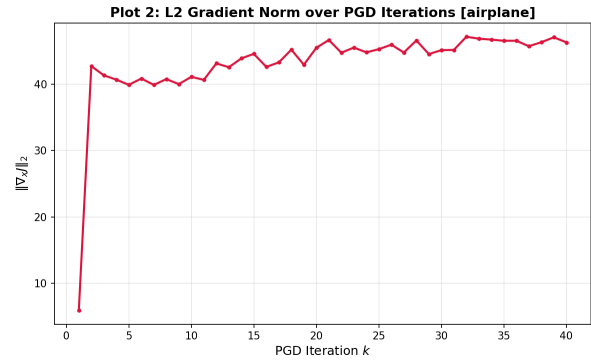


Figure 3: L_2 Norm of the gradient tracking PGD convergence.

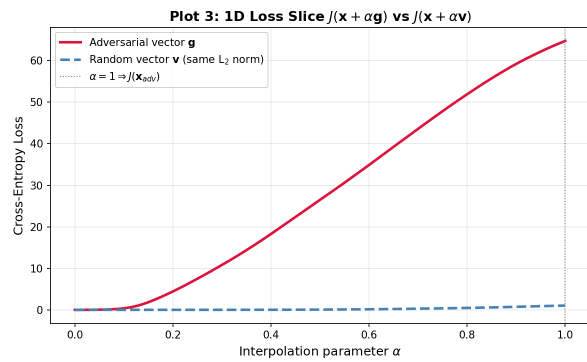


Figure 4: 1D Loss Slice: $J(\mathbf{x} + \alpha \mathbf{g})$ versus $J(\mathbf{x} + \alpha \mathbf{v})$.

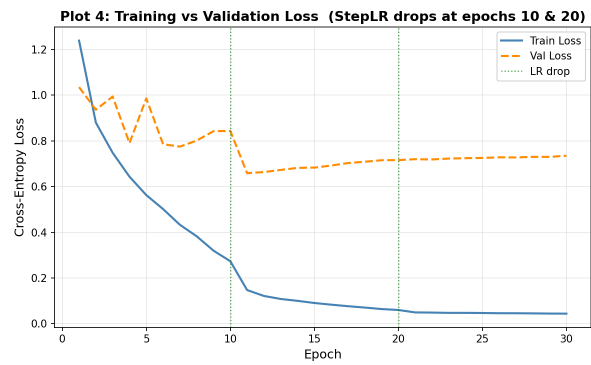


Figure 5: Training and Validation Loss Curve. Inflection points at epochs 10 and 20 reflect StepLR learning rate reductions.

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Plot 5: Visual Triplets — PGD Attack ($\epsilon = 0.05$)

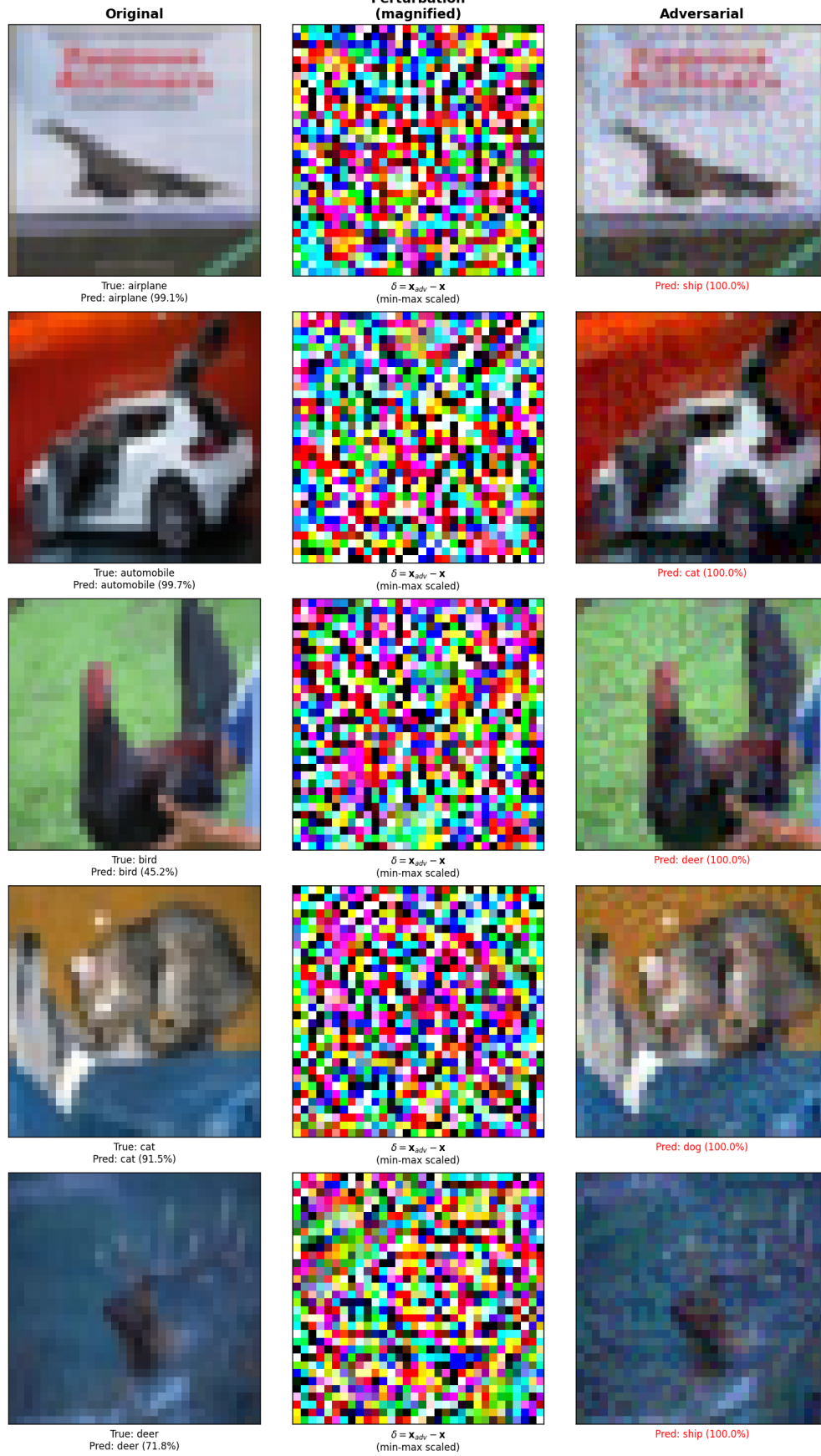


Figure 6: Visual Triplet (5×3 grid): Original Image (Left), Magnified L_∞ Perturbation (Centre), Adversarial Image (Right).